Predicting Beta Lactam Cross reactivity using molecular descriptors

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Clinical Implications

Intelligent application of chemoinformatics algorithms can predict β-lactam cross reactivity with a high degree of sensitivity and specificity. These predictions may be used to fill gaps in empirical data or help adjudicate where there is disharmony in the published literature.

A Recent publication collating cross reactivity data for 1275 combinations of β-lactam antibiotics[1] found 272 instances of clinically relevant 'disharmony' between source authors, 356 situations where there was no data on cross reactivity between a pair of antibiotics, and 1104 situations where there was only one source of information.

We proposed that greater similarity between the R side chains of a pair of β-lactam antibiotics should correlate with a greater risk of immune cross reactivity. This concept was briefly investigated by Picard et al[2], However findings were limited based on a smaller dataset, and use of only a single chemical descriptor.

Since 'similarity' between two molecules would vary depending on what criteria is used, a model was designed combining the scores of multiple different chemoinformatics algorithms, each utilising a different method of generating a chemical descriptor to compare similarity between any pair of β-lactam side chains.

Empirical data on cross reactivity between 1275 combinations of β-lactam antibiotics collated by Hutten et al[1] was transcribed to a spreadsheet and used as the validation for the model. The chemical structure of each of these antibiotics were retrieved from the Reaxys® database[3], split into their component parts (Side chains, β-lactam ring, and fused ring) and stored in SMILES format. A workflow using Python and RDKit[4] was used to generate molecular descriptors using various methods; Substructure Match SMILES, Substructure Match SMARTS, Morgan coefficient, Pharmacophore, Atom Pairs, MACCS, and Maximum common substructure. With these models, either a topological graph, a bit vector, or a count vector is generated from the R side chains of each β-lactam, which are then compared to a second β-lactam's R side chain, and a similarity score between the pair is generated. A pair of side chains is predicted to be cross reactive if its similarity score exceeds a pre-set threshold for an individual algorithm. Results are additive, meaning that if the similarity score of a pair exceeds the threshold of one or more algorithms, then overall, the pair is predicted to be cross reactive. If the similarity score fails to exceed the threshold of at least one algorithm, a pair is predicted to not be cross reactive. The results were collated into a single chart in the same fashion as presented by Hutten et al[1]. Combined performance of the algorithms are validated using the empirical data provided by Hutten et al. For the purposes of validation, results presented by Hutten et al where there is no discrepancy between the source data are considered to be true. Thresholds for each individual algorithm were calibrated to maximise agreement with results presented by Hutten et al where there was no discrepancy between source data on potential cross reactivity and no cross reactivity. With model performance measured by calculating a Youden's J from sensitivity and specificity. Measurements of sensitivity and specificity were agnostic to molecular pairs where there was either no data available, or discrepancy between source data, effectively resulting in the model making 'predictions' for these pairs. Performance was ultimately maximised by invoking a differential evolution algorithm[?] to find the most optimal threshold for each algorithm

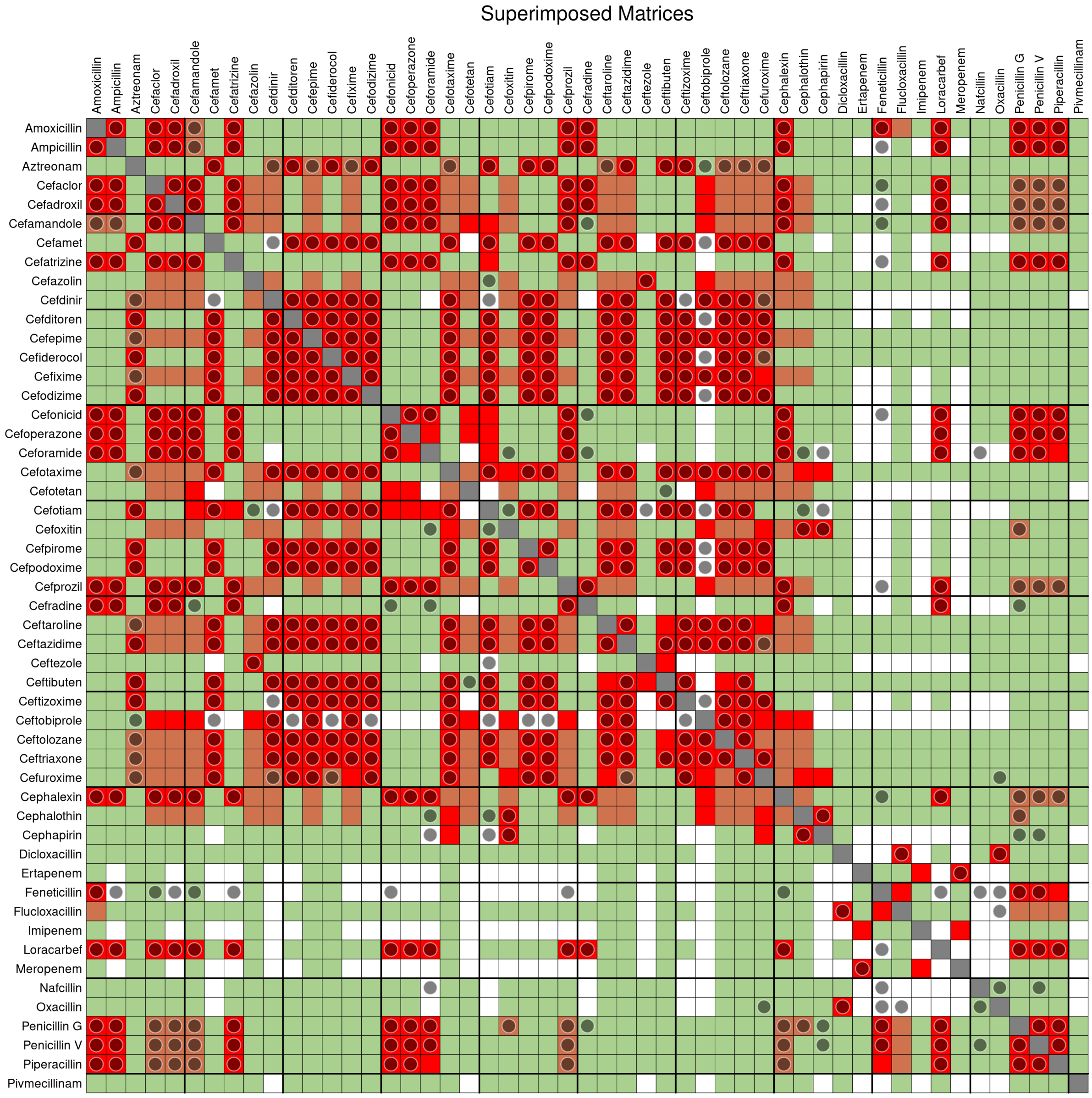

Figure 1: Algorithim predictions superimposed over Hutten et al. Matrix. Note: white:no data; Grey:cross-tabulation similar; Green:no crossreactivity; Safe; Orange:Cinically relevant discrepancy between studies; Red:(potential) crossreactivity: unsafe; Grey circle: Cross reactivity predicted by algorithm.

Table 1: Compared performance of each algorithm Morgan Tanimoto not included in combined result. DM:Direct Matching, MD:Morgan-Dice, PDM:Permissive Direct Matching, Pharmaco:Pharmacophore, MCS:Maximum Common Substructure, AP:Atom Pairs.

Metric	DM	MD	PDM	Pharmaco	MCS	MACCS	AP	Combo
Threshold	N/A	0.4867	N/A	0.67	0.8914	0.7041	0.4709	N/A
Possible Hits	297	297	297	297	297	297	297	297
Total Hits	168	202	183	94	240	270	268	335
True +ve	152	178	161	81	209	221	217	262
False +ve	0	3	1.0	0	1	9	15	19
False -ve	145	119	136	216	88	76	80	35
no Data	4	8	4.0	8	12	20	17	24
Sensitivity	0.512	0.599	0.542	0.273	0.704	0.744	0.731	0.882
Specificity	1	0.995	0.998	1.0	0.998	0.89	0.986	0.971
Youden's J	0.512	0.595	0.541	0.273	0.702	0.731	0.708	0.854

The results of the model as presented in Figure 1 are represented as dots superimposed over a reproduction of the figure from Hutten et al for predicted cross reactivity. As shown in Table 1 the model shows a high level of sensitivity (0.882), and very high specificity (0.971). The model makes 24 predictions of cross reactivity for which there exists no empirical data, and 30 predictions of cross-reactivity where there is disharmony between authors.

Of these cross reactivity predictions, there is a range of agreement between algorithms from full agreement, partial, down to a single algorithm predicting, which represents a theoretical 'confidence' in the result. As more than one algorithm flagging positive means a greater likelihood that a similarity is not artefactual, and represents something significant in the similarity between two molecules.

To our knowledge, since 2019[2] no further published literature investigating the capability of chemoinformatics algorithms to characterise structural similarity between β-lactam side chains and relate this to cross-reactivity, and none that have been robust enough to make formal predictions, or to support empirical data in cases where it exists, but remains in dispute.

While chemoinformatics is a well established field, used heavily in drug discovery and &&&, its use in this application still remains relatively unexplored. This model, while performing impressively&&&, is unlikely to represent the most optimal possible method of characterising and comparing β-lactam side chains and predicting cross reactivity. Further experimentation and validation in this field is likely to yield even more precise and reliable models. With that reliability and would come utility in acute situations where alternative β-lactam antibiotics are required to treat a patient with a recorded allergy.

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